LAN Talk: Simple Two-User Chat Application

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1. INTRODUCTION

LAN Talk is a lightweight, real-time-capable web chat application designed to allow two or more devices on the same Wi-Fi network or Local Area Network (LAN) to exchange messages seamlessly. Built exclusively using **PHP, HTML, and CSS**, the system runs on the Apache web server provided by XAMPP. Messages are persisted in a flat text ledger (`messages.txt`) without requiring SQL databases, JavaScript, AJAX polling, WebSockets, or third-party libraries.

2. FILES USED & ARCHITECTURE

File Name	Functional Role & Description
`index.php`	Landing page that captures the user's handle and securely stores it in a PHP session (`$_SESSION['user']`).
`chat.php`	Main room interface; appends new messages to `messages.txt` , prevents double submissions, and displays conversation threads.
`logout.php`	Destroys active session variables (`session_destroy()`) and redirects back to the login portal.
`messages.txt`	Shared flat-file datastore storing serialized chat records, formatted as one line per message (`Sender|Message`).
`style.css`	Defines card layouts, distinct chat bubble styles (mine vs. theirs), and responsive mobile inputs.

3. WORKING MECHANISM & DATA FLOW

The user enters their handle on `index.php` , which is saved in a PHP session. On `chat.php` , the user types a message and submits it via an HTML POST form. PHP acquires an exclusive file lock (`LOCK_EX`) and atomically appends the entry to `messages.txt` . To prevent duplicate submissions on page refresh, the script implements the **Post/Redirect/Get (PRG)** pattern. The chat page loads all stored lines and renders them safely using `htmlspecialchars()` to defend against Cross-Site Scripting (XSS).

4. RUNNING THE APPLICATION ON HOST LAPTOP

1. Copy the project folder into your Apache document root: `C:\xampp\htdocs\chat-app` .
2. Open the **XAMPP Control Panel** and click **Start** next to Apache.
3. On the server laptop, open a web browser and navigate to `http://localhost/chat-app/` .

5. CONNECTING FROM A SECOND LAPTOP / MOBILE DEVICE

1. On the server laptop, run `ipconfig` in Command Prompt and note the IPv4 Address (e.g., `192.168.1.15`).
2. Connect the second laptop or mobile phone to the same Wi-Fi network.
3. On the second device, open a browser and navigate to `http://192.168.1.15/chat-app/` (allow Apache through Windows Firewall if prompted).

6. CONCLUSION

LAN Talk proves that stateful, multi-user networked communication can be achieved with simplicity using native PHP sessions, HTTP forms, and flat-file I/O over a standard local Wi-Fi connection.